

Floating point exception in torch.nn.functional.ctc_loss #150835

Closed

vwrewsge opened on Apr 8, 2025 · edited by pytorch-bot

Edits · ···

Describe the bug

When calling `ctc_loss` with empty tensors on CUDA (device="cuda"), a Floating point exception occurs. This does not happen on CPU, where a proper error is raised.

To reproduce

```
import torch
import torch.nn.functional as F
device = "cuda" # "cpu" is fine
num_classes = 4
log_probs = torch.rand(0, 0, num_classes, device=device)
targets = torch.tensor([], device=device, dtype=torch.long)
input_lengths = torch.tensor([], device=device, dtype=torch.long)
target_lengths = torch.tensor([], device=device, dtype=torch.long)
result = F.ctc_loss(log_probs, targets, input_lengths, target_lengths, reduction='none')
```

Output

```
Floating point exception (core dumped)
```

Versions

PyTorch version: 2.6.0+cu124

Is debug build: False

CUDA used to build PyTorch: 12.4

ROCM used to build PyTorch: N/A

cc @albanD @mruberry @jbschlosser @walterddr @mikaylagawarecki @ptrblck @msaroufim @eqy

 **bdhirsh** added **module: nn** **triaged** **module: edge cases** on Apr 9, 2025

 **albanD** added **module: cuda** on Apr 9, 2025

 **zeshengzong** mentioned this on Apr 10, 2025

 Add check for ctc_loss targets param #150981

 `torch.nn.functional.ctc_loss` inconsistent implementation and docs #150995

  **pytorchmergebot** closed this as completed in [01f226b](#) on Apr 14, 2025

  **jianyizh** added a commit that references this issue on Apr 14, 2025

Add check for ctc_loss targets param ([pytorch#150981](#)) 

 [01f226b](#)

  **timocafe** added a commit that references this issue on Apr 16, 2025

Add check for ctc_loss targets param ([pytorch#150981](#)) 

 [46fc5d8](#)

  **amathewc** added a commit that references this issue on Apr 17, 2025

Add check for ctc_loss targets param ([pytorch#150981](#)) 

 [cb2358e](#)

 **andyedwardsibm** on Apr 25, 2025 

FYI this has been assigned [CVE-2025-3730](#)

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 **songohannyc** on May 21, 2025 

Will this be in 2.6.1 and 2.7.1? Is it currently blocked by [#150981](#) ?

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 **michaelfarrell76** on Jun 3, 2025 

hi! is there an estimated time for when 2.7.1 will be released with this vuln cleared?

 **MattTheCuber** on Jul 12, 2025 

Guess this in 2.7.1 as well.

 **iloveskywsd** on Jul 18, 2025 

[@albanD](#) [@zeshengzong](#) Do you know when will this fix be included?

eqy on Jul 18, 2025

Collaborator ...

[@iloveskywsd](#) it should be in current nightlies as well as 2.8

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iloveskywsd on Aug 13, 2025

...

[@iloveskywsd](#) it should be in current nightlies as well as 2.8

[@eqy](#) I just check the release note for 2.8.0. This fix wasn't mentioned in it. What is the current status on this?

mikaylagawarecki on Aug 13, 2025

Contributor ...

I'm sorry, I think it might have been categorized as not user facing and hence wasn't in the release notes, but it should be in 2.8 as mentioned above

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szuliq mentioned this [on Aug 28, 2025](#)



[\[GHSA-887c-mr87-cxwp\] PyTorch Improper Resource Shutdown or Release vulnerability github/advisory-database#6052](#)



depthfirst-dev mentioned this in 4 pull requests [on Oct 24, 2025](#)



[\[depthfirst-5905\] Upgrade multiple packages in uv.lock seshanktest/tts#1](#)



[\[depthfirst-5909\] Upgrade multiple packages in uv.lock seshanktest/tts#2](#)



[\[depthfirst-5911\] Upgrade multiple packages in uv.lock seshanktest/tts#3](#)



[\[depthfirst-5913\] Upgrade multiple packages in uv.lock seshanktest/tts#4](#)

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Assignees

No one assigned

Labels

module: cuda

module: edge cases

module: nn

triaged

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Code with agent mode

▼

 **Add check for ctc_loss targets param**
pytorch/pytorch

Participants



+5